WFormer: A Transformer-Based Soft Fusion Model for Robust Image Watermarking

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***Abstract*—Most deep neural network (DNN) based image watermarking models often employ the encoder-noise-decoder structure, in which watermark is simply duplicated for expansion and then directly fused with image features to produce the encoded image. However, simple duplication will generate watermark over-redundancies, and the communication between the cover image and watermark in different domains is lacking in image feature extraction and direction fusion, which degrades the watermarking performance. To solve those drawbacks, this paper proposes a Transformer-based soft fusion model for robust image watermarking, namely WFormer. Specifically, to expand watermark effectively, a watermark preprocess module (WPM) is designed with Transformers to extract valid and expanded watermark features by computing its self-attention. Then, to replace direct fusion, a soft fusion module (SFM) is deployed to integrate Transformers into image fusion with watermark by mining their long-range correlations. Precisely, self-attention is computed to extract their own latent features, and meanwhile, cross-attention is learned for bridging their gap to embed watermark effectively. In addition, a feature enhancement module (FEM) builds communication between the cover image and watermark by capturing their cross-feature dependencies, which tunes image features in accordance with watermark features for better fusion. Experimental results show that the proposed WFormer outperforms the existing state-of-the-art watermarking models in terms of invisibility, robustness, and embedding capacity. Furthermore, ablation results prove the effectiveness of the WPM, the FEM, and the SFM.**

***Index Terms*—Watermarking, transformer, soft fusion, cross-attention.**

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I. Introduction

With the development of generative artificial intelligence (AI), more and more digital images are being generated for different purposes, which are considered as the digital property [1], [2]. When these generated images are transmitted on public Internet platforms (e.g., Twitter, Facebook, WeChat, etc.), they may be maliciously tampered (e.g., compressed, cropped, scaled, etc.) or copied by some illegal users, resulting in a copyright threat. Thus, protecting the copyright of those images generated by the AI models is of the real-world application. Robust watermarking technology is the key solution to protect copyright or trace information leakage by embedding the secret message into the multimedia [3], [4]. Owing to the easy acquirement and wide spread of digital images, the robust image watermarking embeds watermark into the image invisibly, and hopes to recover watermark completely from the watermarked image even if it is distorted [5], [6]. The image quality and watermarking robustness are two important requirements for robust image watermarking [7], [8].

In the past years, many traditional robust watermarking methods based on frequency domain were presented, in which latent image features are tried to be mined for hiding watermark [9], [10], [11]. However, those image features are extracted manually and cannot represent the main energy for various images, and consequently, it may lead to poor robustness. Since deep neural network (DNN) can be trained to learn deep image features from different image datasets, a few watermarking models based on convolutional neural network (CNN) were proposed to attain high watermarking performance [12], [13], [14], [15], [16], [17], [18], [19], [20], [21]. The existing CNN based watermarking models often employ an encoder-noise-decoder (END) structure as the main backbone. The encoder fuses watermark with the image, and oppositely the decoder tries to recover watermark. Meanwhile, the noise layer is used to train different noises for robustness. In the END structure, the encoder and the decoder affect each other, that is, the encoded image determines how robust watermark, and the accuracy of watermark extraction influences the encoding process in turn. In fact, the decoder is only the inverse process of the encoder, and thus, designing an effective encoder is essential to the whole watermarking performance. The encoder mainly consists of three folds: watermark preprocess, image feature extraction, and watermark fusion with the image feature. At first, watermark preprocess mainly expands watermark over the whole image to increase

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watermark redundancy for robustness. Then, image features are extracted by convolution operations, and finally, image features are fused with the expanded watermark to generate the encoded image.

In detail, for watermark preprocess, watermark is often simply duplicated for generating redundancy before fusion so that it can be distributed over the whole image to increase robustness, such as Hidden [13] and TSDL [15]. However, this kind of operation sometimes cannot produce effective watermark features, and generates over-redundancy inversely, thereby degrading the image quality. In order to solve this drawback, Jia et al. [17] used some convolution operations to expand watermark over the image for redundancy instead of simple duplication. Although more effective and expanded watermark features can be learned for different images to some extent compared with simple duplication, the convolution operation only has a limited receptive field, which leads to incomplete watermark features extracted from the expansion process. In the image feature extraction aspect, we hope that image features are extracted to be in consistent with watermark features in advance before watermark fusion, but conventional CNN based watermarking models often omit watermark features, such as MBRS [17] and DA [18]. We need to build relationships between the cover image and watermark to adjust image features according to watermark features. In the fusion aspect, the expanded watermark is fused with image features to obtain the encoded image, such as Fang's [16] and Ma's [20]. However, image and watermark features are directly fused only by employing channel concatenation without any appropriate guidance or interaction so that the cross-feature dependencies between them are not built. Their long-range relations result in poor fusion performance, since watermark features are not adjusted toward image detail or structure. Overall, in the traditional CNN based watermarking models, the above aspects of the encoder are not designed properly without mining latent watermark features and considering correlations between the image and watermark, which limits the watermarking performance.

Recently, Transformers have been successfully applied in computer vision and natural language tasks, because of their effective representation learning [22], [23], [24], [25]. For instance, Transformers can compute self-attention to achieve latent image features and calculate cross-attention to construct the long-range interaction between two different modalities [26], [27], [28]. Inspired by the advantages of Transformers, we hope to integrate its architecture into watermarking network models to address the above mentioned drawbacks of traditional CNN based watermarking models. However, if Transformers are directly applied into existing watermarking frameworks, long-range relationships between the image and watermark may be missed for tuning image features in accordance with watermark. Furthermore, how to use Transformers to fuse the image with watermark by establishing their correlations still needs to be studied. Thus, designing a proper Transformer based watermarking model for obtaining superior performance is still challenge. Up to now, Transformer based watermarking models are rarely reported.

In this paper, we propose a soft fusion network based on Transformer for robust image watermarking, namely WFormer. To overcome the drawback of simple watermark duplication, a watermark preprocess module (WPM) based on Transformer is deployed to extend the watermark through automatic learning with self-attention, so that valid watermark features are extracted to avoid watermark over-redundancy. To extract latent image features for decreasing useless ones, a feature enhancement module (FEM) is presented to adjust image features in concordance with watermark features by capturing their cross-feature dependencies. More critically, to figure out a solution to channel concatenation fusion, a soft fusion module (SFM) is designed by computing mixed attentions. Specifically, self-attention is computed to extract latent features of the image and watermark, respectively, and meanwhile, cross-attention is calculated to allow their communication to embed watermark in terms of image quality and robustness. Experimental results show that the proposed WFormer can resist most image attacks while obtaining the high quality of the encoded image.

A. Contributions

The main contributions of this paper are listed as follows.

- We investigate the limitation of direct fusion in the existing CNN based watermarking framework, and thus propose the SFM based on Transformer to bridge the gap between the image and watermark by computing their own self-attentions as well as cross-attentions. As a result, watermark is embedded effectively, and the image quality and watermarking robustness are increased.
- We discover that the image features extracted by convolution operations without considering watermark information include useless ones for embedding watermark, which degrades the watermark fusion performance. For this reason, the FEM tries to tune image features consistent with watermark features in advance by acquiring cross-feature dependencies between them to help obtain better watermark fusion.
- Although simple duplication can expand watermark over the whole image for watermarking robustness, the expanded watermark is often over-redundant for different images. The WPM based on Transformer is deployed to expand watermark by automatic learning with self-attention computation, so that valid watermark features are produced to be embedded.

B. Organization

The remainder of this paper is organized as follows. Section II discusses the related work of the proposed watermarking model. Section III introduces the proposed watermarking model based on Transformer. In Section IV, experimental results are tested to be analyzed. Section V concludes this paper.

II. Related Work

A. Robust Watermarking

In the past years, many traditional robust watermarking methods were presented mainly based on decomposition domains, such as discrete Fourier transform (DFT) [9], discrete cosine transform (DCT) [10], and discrete wavelet transform (DWT)

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TABLE I
ADVANTAGES AND DISADVANTAGES OF EXISTING CNN-BASED WATERMARKING MODELS

Model	Advantages	Disadvantages
HiDDeN [13]	• First applying an end-to-end training approach to the watermarking field • Improved robustness by introducing a noise layer • Proposing differentiable JPEG compression	• Poor invisibility and robustness • Small embedding capacity
ReDMark [14]	• Watermark can be embedded in any transform domain • Improved security and robustness especially against Crop • Proposing new differentiable JPEG compression	• Poor robustness against untrained attacks • Needing to build differentiable model for non-differentiable attacks
TSDL [15]	• Robust to some black-box attacks • The encoder can be trained with arbitrary attacks	• Poor robustness against serious distortions
Fang et al. [16]	• Robustness against various black-box distortions • Robustness against common non-differentiable distortions • High imperceptibility	• Poor robustness against Crop and Rotate attacks • Training is time-consuming
MBRS [17]	• Robustness against JPEG compression • High imperceptibility	• Small embedding capacity • Extracting image features without considering watermark features • Fusion between the image and watermark without interaction
DA [18]	• Training without introducing distortions	• Poor robustness against various distortions
Zhong et al. [19]	• Model training is unsupervised • Robust against cutting-edge and challenging camera applications attacks	• Poor robustness against geometric and perspective distortions
CIN [20]	• Robust against specific distortions • High imperceptibility	• Poor robustness against combined distortions
De-END [21]	• Efficient coupling of the encoder and the decoder • High robustness against specific distortions	• Cannot work against combined distortions

[11]. Hsu et al. [29] proposed a cross-DCT based watermarking method combining mean modulation and mixed modulation, which effectively improves watermarking imperceptibility and robustness. However, the method uses a fixed embedding strength for all blocks and does not consider the humane visual system. In order to solve the above problems, Li et al. [30] proposed a visual saliency-guided watermarking method, which adjusts the embedding strength of different image blocks according to attentive information. Choudhary et al. [31] proposed a two level DWT-based watermarking method, which embeds the whole image as watermark into the cover image considering the visibility factor. Consequently the high visual quality is obtained, but robustness is not high. Thus, frequency domain based watermarking methods have the limitation in obtaining the balance between invisibility and robustness.

With the development of deep neural network (DNN), a few convolutional neural network (CNN) based watermarking models have been presented as shown in Table I, wherein advantages and shortcomings are described. Zhu et al. [13] designed HiDDeN, which is an auto-encoder based watermarking network. Via joint training of the encoder-decoder as well as the differentiable noise layer, the encoded image is obtained, which has the ability to resist common image processing. Ahmadi et al. [14] introduced an adaptive diffusion watermarking framework, namely ReDMark, which supports different required transforms to embed watermark. The residual structure of the encoder with adjustable parameters is used to control the watermarking strength. However, to make the gradient propagated back to the encoder, a differentiable model should be established for non-differentiable distortion, such as JPEG compression. In order to solve this limitation, Liu et al. [15] deployed a two-stage separable deep learning network (TSDL). In the first stage, the encoder and the decoder are trained initially without the noise layer. In the second stage, the parameters of the encoder are preserved and the decoder is trained again by real JPEG compression. Nevertheless, since the encoder is not enhanced again and only the decoder is tuned for robustness, watermark cannot be accurately extracted when the watermarked image is under heavy distortions. In order to improve watermarking robustness, Fang et al. [16] presented triple-stage training based watermarking framework. Different from TSDL, one extra stage is added to enhance encoded features by using a mask-guided enhanced technique. Nevertheless, this kind of stage-wise training still cannot make the encoder get information about decoding with JPEG compression, which limits corresponding robustness. To improve resistance to JPEG compression, Jia et al. [17] utilized mini-batch of real and simulated JPEG compression (MBRS) for training. Luo et al. [18] presented a watermarking model network based on distortion agnostics (DA), and used channel coding and adversarial training to increase watermark redundancy for robustness, which has the capability of resisting some trained and untrained attacks. Zhong et al. [19] realized the ability to resist different image attacks by sparsely activating neural connections with watermarking related regions in the redundant invariant layer of the network. Ma et al. [20] combined invertible and non-invertible (CIN) mechanisms to achieve watermarking imperceptibility and robustness. The above watermarking models employ the END structure, yet the encoder often embeds many redundant features that are unnecessary for the decoder, since the encoder and the decoder are not satisfactory coupling. To solve this drawback, Fang et al. [21] presented a decoder-encoder-noise-decoder (De-END) structure, where the image is analyzed by the decoder at first for coupling with the encoder, and consequently redundant features will be decreased to increase the watermarking performance.

B. *Vision Transformers*

The Transformer architecture has been considered to replace the dominant position of CNN due to its excellent performance,

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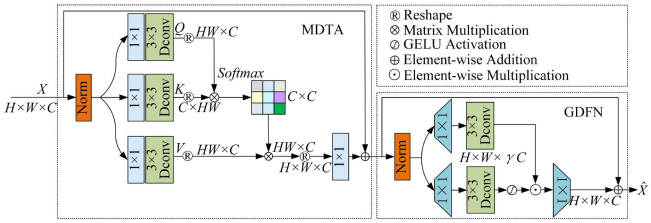


Fig. 1. Structure of Transformer Block

which is deployed for natural language processing at first [22]. Later, Vision Transformers are designed and applied in computer vision tasks, such as image recognition [23], [32], segmentation [33], [34], object detection [35], [36], and so on. Vision Transformers can decompose the image into a sequence of tokens, and then obtain their mutual relationships. The attentive features of Vision Transformers can be adaptive to the given content by learning long-range dependencies among image tokens. However, in Transformers, the computation complexity of self-attention is increased dramatically when the image token number is raised, which leads to poor application in high-resolution images. To alleviate the computation complexity, self-attention is calculated within local image regions, such as Swin Transformer [27]. Moreover, above Transformers still have the shortcoming of global context aggregation. To overcome this limitation, Zamir et al. [37] presented a novel Transformer model named Restormer, which can learn long-range dependencies while preserving computational efficiency. The core component of Restormer is the Transformer Block, which consists of multi-Dconv head transposed attention (MDTA) and Gated-Dconv feed-forward network (GDFN) as illustrated in Fig. 1. Specifically, in MDTA, for a given $X \in \mathbb{R}^{H\times W\times C}$, normalization, 1×1 convolution, and 3×3 depth-wise convolution operations are applied to obtain Query (Q), Key (K) and Value (V) 3D tensors, respectively, wherein the 1×1 convolution operation is used to gather the pixel-wise cross-channel context, and the 3×3 depth-wise convolution operation is utilized to aggregate the channel-wise spatial context. Then, those 3D tensors are reshaped to 2D tensors, respectively, which are used to compute multi-headed self-attention. In GDFN, the layer normalization operation is used to generate two parallel paths. These two paths are operated separately and identically with 1×1 convolution and 3×3 depth-wise convolution operations, and only the first path needs to be activated with the GELU non-linearity [38]. After convolution operations, feature maps of two paths are element-wise multiplied, and finally, the output feature map $\hat{X} \in \mathbb{R}^{H\times W\times C}$ is obtained via the residual connection.

To our knowledge, Transformers are not applied in watermarking up to now. We will make full use of their advantages, and integrate Transformer Blocks into the proposed watermarking model for improving the watermarking performance.

III. The Proposed Watermarking Model Based on Transformer

In this paper, a soft fusion model based on Transformer for robust image watermarking, namely WFormer, is proposed. In the following, the whole framework of WFormer is depicted, which includes the encoder, the decoder, the noise layer, and the discriminator as illustrated in Fig. 2. Then, three main modules the watermark preprocess module (WPM), the feature enhancement module (FEM), and the soft fusion module (SFM) are expounded in detail, respectively.

A. Overview of WFormer

Let original watermark be $W_o \in \{0,1\}^L$ with the length of L, and the cover image be $I_c \in \mathbb{R}^{H\times W\times 3}$, where $H\times W$ is the spatial dimension. As illustrated in Fig. 2, the proposed WFormer employs the encoder-noise-decoder (END) structure, and the whole framework consists of four parts: the encoder, the decoder, the noise layer, and the discriminator. The encoder E with the parameter θ_e is used to fuse I_c with W_o to generate the encoded image (watermarked image) I_e. In the noise layer N, different noises are randomly added into I_e to generate the noised image I_n for robustness training. The decoder D with the parameter θ_d is employed to recover watermark from I_n. The discriminator A with the parameter θ_a is used to give feedback on whether I_e is close to I_c. Each part is expounded in detail as follows.

Encoder E: The goal of E is to embed W_o into I_c with low distortion and to have the capability of resistance to different image attacks. E is mainly consisting of the WPM, the FEM, and the SFM. The WPM is to expand watermark with proper redundancy for supporting the encoding process by employing Transformers to compute self-attention. The FEM is utilized to tune image features consistent with watermark features by capturing their cross-feature dependencies. The SFM is used to bridge the gap between the image and watermark by mining their self-attention and cross-attention features. Specifically, at first, the WPM is used to expand W_o to obtain the feature map, denoted as $F_w \in \mathbb{R}^{H\times W\times C}$, wherein C is the channel number. In the meantime, the cover image I_c is operated by the 3×3 convolution operation to generate the initial image feature map, denoted as $F_c \in \mathbb{R}^{H\times W\times C}$, which cannot represent the main energies. Then, F_c and F_w are fed to some FEMs to generate the enhanced feature map, denoted as $F_f \in \mathbb{R}^{H\times W\times C}$, wherein the number of FEMs is K_1. F_f and F_w are input to the SFM for fusing F_f with F_w to generate the watermarked feature map, named as $F_s \in \mathbb{R}^{H\times W\times C}$. Finally, F_s is operated by the Transformer Block to be connected with I_c by the skip connection, and then input into the 3×3 convolution layer to produce the encoded image $I_e \in \mathbb{R}^{H\times W\times 3}$.

The aim of encoder training is to make I_c approach I_e by updating the parameter θ_e, following the loss L_E:

$$L_E = MSE\left(I_c, I_e\right) = MSE\left(I_c, E_p\left(\theta_e, I_c, W_o\right)\right), \quad (1)$$

where $E_p(\cdot)$ is the encoding process and $MSE(\cdot)$ computes the mean square error.

Noise Layer N: The target of N is to increase the ability to resist different image attacks. In N, different differentiable image processing operations are added into I_e to generate the noised image I_n, which will be input to the decoder for the training of watermark extraction. The trained noises are expected

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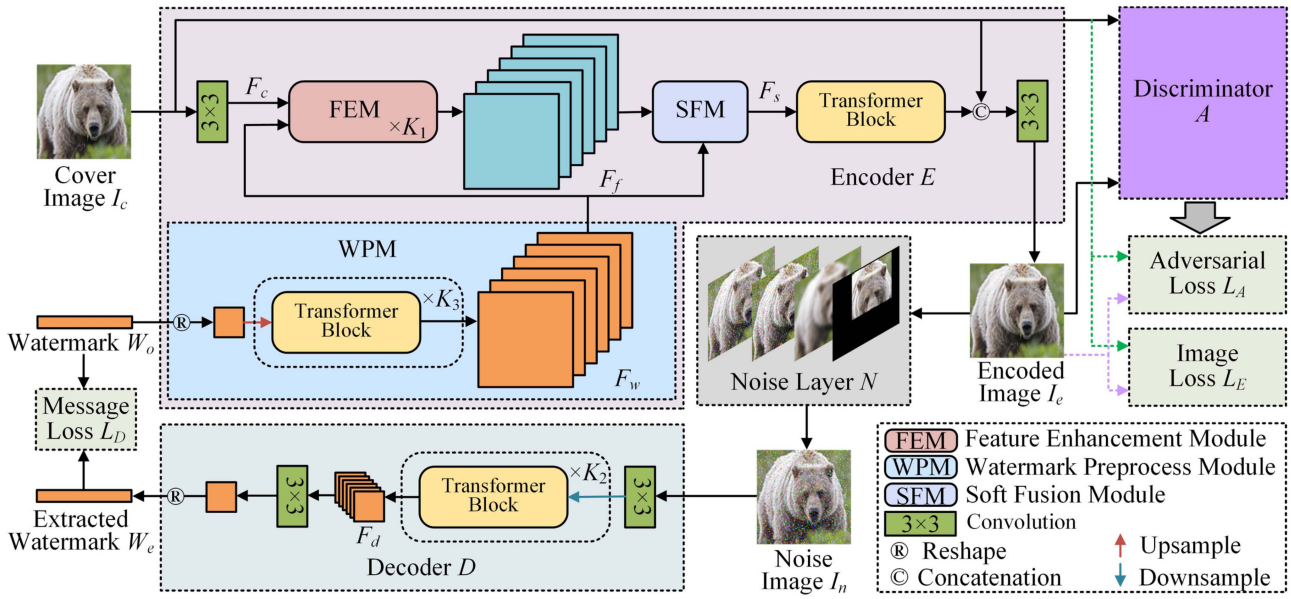


Fig. 2. Framework of WFormer

to represent the nature of various noises, such as JPEG Compression, Crop, Median filtering, and so on.

Decoder D: The objective of D is to extract watermark completely from the noised image I_n. At first, I_n is operated by the 3×3 convolution operation, and then is processed by pixel-unshuffle [39] and Transformer Block operations to produce $F_d \in \mathbb{R}^{H \times W \times C}$, wherein the number of down-sampling operations is K_2. K_2 is determined by the number of up-sampling operations in the encoder, and introduced in Part B of Section III. Finally, F_d is converted to one dimensional channel by the 3×3 convolution operation, and reshaped to extract watermark $W_e \in \{0,1\}^L$.

The purpose of decoder training is to minimize the distance between W_o and W_e by updating the parameter θ_d, following the loss L_D:

$$L_D = MSE\left(W_o, W_e\right) = MSE\left(W_o, D_p\left(\theta_d, I_n\right)\right), \quad (2)$$

where $D_p(\cdot)$ means the decoding process.

Discriminator A: The essential purpose of A is to distinguish I_e from I_c for increasing the quality of I_e. This process is similar to Conditional-GAN [40], which is used to generate the image satisfying a given condition. A is mainly consisting of four convolutions and one average pooling operation. As the adversary of E, A tries to identify that I_e is fake by updating θ_a

$$L_{Dis} = \log\left(1 - A_p\left(\theta_a, I_e\right)\right) + \log\left(A_p\left(\theta_a, I_c\right)\right), \quad (3)$$

where $A_p(\cdot)$ is the discriminating process, and L_{Dis} is the training loss for A.

On the other hand, θ_e is updated in the encoder to generate a similar image as I_c, following the loss L_A:

$$\begin{aligned} L_A &= \log\left(A_p\left(\theta_a, I_e\right)\right) \\ &= \log\left(A_p\left(\theta_a, E_p\left(\theta_e, I_c, W_o\right)\right)\right), \end{aligned} \quad (4)$$

The final loss of WFormer for co-training the encoder-decoder is consisting of the image quality loss, the adversarial loss, and the decoding loss, and can be expressed as:

$$L_t = \lambda_1 L_E + \lambda_2 L_A + \lambda_3 L_D. \quad (5)$$

where λ_1, λ_2, and λ_3 are weight factors.

B. Watermark Preprocess Module (WPM)

The main purpose of the WPM is to expand W_o over the whole image for resisting different image attacks. In order to generate an expanded and valid watermark feature map F_w without over-redundancy, the Transformer based WPM is deployed.

To be specific, watermark is reshaped into the matrix with the size of $h \times w$, wherein $h \times w = L$. Then, the matrix is amplified for up-sampling by using pixel-shuffle [39], and effective features $F_p \in \mathbb{R}^{H \times W \times C}$ are extracted by computing self-attention via the Transformer Block operation. Repeat those operations until F_w is obtained for the encoder E. During the handling of watermark expansion, each pixel-shuffle operation can double the width and height of the input tensor. In general, the watermark length L and the image size $H \times W$ are required to obey the following equation:

$$L = (h \times w) = \left(H/2^{K_3}\right) \times \left(W/2^{K_3}\right). \quad (6)$$

where K_3 is the iterative times of the pixel-shuffle operation and is determined by L, H, and W. For example, when $L = 64$ and $H \times W = 128 \times 128$, K_3 is 4. It is worth noting that the parameter K_2 in the decoder should be the same as K_3 for accurate decoding. It is mainly because K_2 and K_3 denote the number of the up-sampling operations in the encoder and the number of down-sampling operations in the decoder, respectively, and they should be symmetry.

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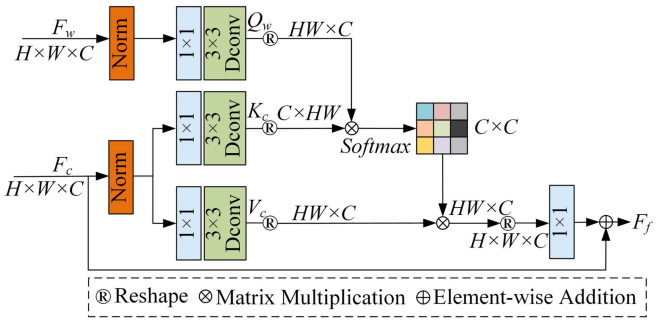


Fig. 3. Structure of FEM

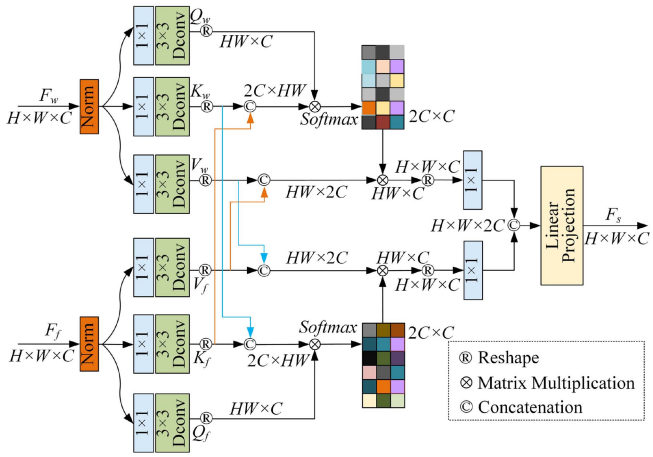


Fig. 4. Structure of SFM

For computing the Transformer Block as illustrated in Fig. 1, the input watermark features are projected into Query (Q_p), Key (K_p) and Value (V_p), respectively. Then, Q_p and K_p are multiplied to obtain the attention map with the size of $C \times C$. This attention map represents the global correlation between watermark feature channels. In the following, V_p and the attention map are multiplied to obtain the output watermark features, wherein large weights are assigned to watermark feature channels for increasing watermarking performance, and small weights are given to useless or redundant watermark features.

C. Feature Enhancement Module (FEM)

The FEM aims to extract effective latent image features of the cover image. As illustrated in Fig. 3, the FEM is designed to enhance image features in accordance with watermark features by capturing their cross-feature dependencies. It permits communication between the image and watermark to attain their feature correlations for tuning latent image features by computing cross-attention. Meanwhile, some redundant image features having low correlations with watermark are suppressed for watermark embedding.

Formally, the watermark feature F_w is operated by using the layer normalization [41], the 1×1 convolution, and the 3×3 depth-wise convolution operations to generate Query (Q_w). Here, before computing the covariance of features to produce global attention, layer normalization is utilized for projection, and depth-wise convolutions are mainly employed to emphasize the local context. Meanwhile, the image feature map F_c is transformed to obtain Key (K_c) and Value (V_c) by using the same operation, respectively. Q_w, K_c and V_c are three basic elements for computing the cross-attention of communication between the cover image and watermark in the FEM.

Then, Q_w, K_c, and V_c are reshaped to generate 2D tensor feature maps $\hat{Q}_w \in \mathbb{R}^{HW \times C}$, $\hat{K}_c \in \mathbb{R}^{C \times HW}$ and $\hat{V}_c \in \mathbb{R}^{HW \times C}$, respectively. Finally, cross-attention is computed to obtain the output feature of the FEM as

$$F_f = Conv\left(Attention\left(\hat{Q}_w, \hat{K}_c, \hat{V}_c\right)\right) + F_c,$$
$$Attention\left(\hat{Q}_w, \hat{K}_c, \hat{V}_c\right) = \hat{V}_c \cdot Softmax\left(\frac{\hat{K}_c \cdot \hat{Q}_w}{\sqrt{C}}\right). \quad (7)$$

where $Conv(\cdot)$ is the convolution operation, and $Attention(\cdot)$ computes cross-attention between the image and watermark.

In the computation process of cross-attention, the matrix multiplication of $\hat{Q}_w$ and $\hat{K}_c$ is firstly performed to obtain the cross-attention map, which indicates the correlation degree between image and watermark features. Larger values indicate the closer relationships between the image and watermark, and vice versa. Then, $\hat{V}_c$ and the cross-attention map are multiplied to output the final adjusted image features, in which the image features with high correlations of watermark are enhanced, and the redundant image features are suppressed. After building cross-dependencies, image features are tuned for the subsequent process of watermark fusion.

D. Soft Fusion Module (SFM)

The goal of the SFM is to fuse the cover image with watermark properly. The SFM is designed by making use of their cross-feature dependencies and their own latent features as illustrated in Fig. 4. In essence, a hybrid interaction operation is utilized in the SFM by computing both self-attention and cross-attention on tokens from the cover image and watermark. Self-attention is mainly in charge of extracting their own features to be highlighted, while cross-attention is responsible for building communication between them to bridge their gap.

Formally, the watermark feature F_w is operated by using the layer normalization [41], the 1×1 convolution, and the 3×3 depth-wise convolution operations to generate Query (Q_w), Key (K_w), and Value (V_w), respectively. In the same way, the cover image feature F_f is processed to obtain Q_f, K_f and V_f, respectively. Those 3D tensor features are reshaped to 2D tensor features $\hat{Q}_w \in \mathbb{R}^{HW \times C}$, $\hat{K}_w \in \mathbb{R}^{HW \times C}$, $\hat{V}_w \in \mathbb{R}^{HW \times C}$, $\hat{Q}_f \in \mathbb{R}^{HW \times C}$, $\hat{K}_f \in \mathbb{R}^{HW \times C}$ and $\hat{V}_f \in \mathbb{R}^{HW \times C}$, respectively. Attention maps of watermark and the cover image are computed as

$$A_w = Attention_w\left(\hat{Q}_w, \hat{K}_m, \hat{V}_m\right)$$

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$$= \hat{V}_m \cdot Softmax\left(\frac{\hat{K}_m \cdot \hat{Q}_w}{\sqrt{C}}\right),$$

$$A_c = Attention_c\left(\hat{Q}_f, \hat{K}_m, \hat{V}_m\right)$$

$$= \hat{V}_m \cdot Softmax\left(\frac{\hat{K}_m \cdot \hat{Q}_f}{\sqrt{C}}\right),$$

$$\hat{K}_m = Concat\left(\hat{K}_w, \hat{K}_f\right), \hat{V}_m$$

$$= Concat\left(\hat{V}_w, \hat{V}_f\right), \quad (8)$$

where $Attention_w(\cdot)$ and $Attention_c(\cdot)$ computes the self-attention and cross-attention of watermark and the cover image, respectively, and obtain the feature attention maps A_w and A_c. Then, A_w and A_c are concatenated to generate watermarked features for the final encoded image:

$$F_s = P\left(Concat\left(Conv\left(A_w\right), Conv\left(A_c\right)\right)\right). \quad (9)$$

where $P(\cdot)$ is a linear projection and $Concat(\cdot)$ is the concatenation operation.

In SFM, since $\hat{K}_m$ is concatenated by $\hat{K}_w$ and $\hat{K}_f$, the matrix multiplication of $\hat{Q}_f$ and $\hat{K}_m$ yields a hybrid attention map, which includes both self-attention of the image and cross-attention between the image and watermark. For the self-attention part, the larger values indicate that the corresponding features are more important. For the cross-attention part, the larger values denote that the relationships of the image and watermark are closer. Then, $\hat{V}_m$ and the hybrid attention map are matrix multiplied to obtain A_c, in which effective image features are given high weights to fuse with watermark for high performance. Finally, the watermark features A_w and A_c are concatenated to realize high invisibility and robustness.

IV. Experimental Results and Discussions

Experimental details: The proposed WFormer is implemented by Pytorch [42] and executed on NVIDIA GeForce RTX 3090. All these images are resized to 128 $\times$128 $\times$ 3, and the watermark length $L = 64$. The channel number of the feature map C is set to 64. We use Adam [43] as gradient descent, and its learning rate is set to 10^{-3}. Like common deep learning networks, it employs end-to-end training.

Datasets: The proposed model is trained on COCO [44] dataset and tested on COCO [44], DIV2K [45] and ImageNet [46] datasets. Specifically, for COCO [44] dataset, 10000 images are randomly selected as training dataset and another 5000 images are selected as test dataset. For DIV2K [45], 100 images from the validation set are selected as the test dataset. For ImageNet [46], 5000 images are randomly selected as the test dataset.

Evaluation metrics: We evaluate the proposed model by using watermarking invisibility and robustness, where invisibility is evaluated by peak signal to noise ratio (PSNR) [47] and structural similarity (SSIM) [48], respectively. Bit accuracy (BA) and normalization correlation (NC) are used to evaluate robustness.

TABLE II
Performance of λ_1, λ_3, and λ_2

Criteria		PSNR(dB)	SSIM	BA[.%]
λ_1, λ_3, λ_2	10,10,0.0001	46.69	0.9866	97.42
	5,10,0.0001	44.59	0.9853	98.73
	3,10,0.0001	44.49	0.9802	99.48
	1,10,0.0001	41.27	0.9614	99.81
	3,10,0.0005	45.04	0.9833	98.51
	3,10,0.00005	43.81	0.9761	99.53

Baseline: We evaluate the proposed WFormer by training a specific image attack and combined image attacks, respectively. To prove the effectiveness of the proposed WFormer, the state-of-the-art (SOTA) watermarking models based on DNN (i.e., HiDDen [13], ReDMark [14], TSDL [15], Fang [16], MBRS [17], DA [18], CIN [20] and De-END [21]) are used as baselines for comparisons. Similar as the proposed model, most of them use the END structure for watermarking, such as HiDDen [13], ReDMark [14], and TSDL [15]. Nevertheless, based on END, De-END [21] adds the decoder to analyze the image for coupling with the encoder. To resist noises, they also use different training strategy. Many watermarking models utilize only one stage training, such as HiDDen [13], ReDMark [14] and CIN [20]. However, non-differentiable distortion cannot be trained. To solve this kind of problem, TSDL [15] proposed a two-stage training process, where the noise layer is removed for training in the first stage, and noise training is added in the second stage while parameters of the encoder are preserved. Fang [16] also improved TSDL [15], and a three-stage training watermarking model is designed to resist high-intensity distortion. Since real JPEG compression is the typical non-differentiable distortion, it is simulated in MBRS [17] to increase robustness against JPEG compression, and the proposed model also uses this way. In DA [18], noise training is not required to resist some of these attacks. Thus, above compared models are typical watermarking works considering different structures and training strategies. Furthermore, we have tried experiments of TSDL [15], Fang [16], DA [18], CIN [20], and De-END [21], respectively, and failed to achieve the best performance reported by them. To make fair comparisons, we directly use their published results. In addition, we retrain other watermarking models under the same experimental environment.

A. Parameter Discussion

Parameters of the proposed WFormer are related to the watermarking performance, and in this section, we discuss them. In (5), λ_1 and λ_2 denote the weights of the image quality, i.e., invisibility, and λ_3 emphasizes the watermark extraction accuracy, i.e., robustness. We fix λ_3 and λ_2 to 10 and 0.0001, respectively, and then λ_1 is set to 10, 5, 3 and 1, respectively. From Table II, we find that as λ_1/λ_3 decreases, PSNR and SSIM are reduced, while BA is increased. When λ_1 is changed from 5 to 3, PSNR and SSIM have a slight decrease while BA is improved by 0.75%. However, when λ_1 is changed from 3 to 1, PSNR is decreased by more than 3 dB and BA is only increased by 0.33%. Furthermore, when $\lambda_1 = 3$ and $\lambda_3 = 10$ are fixed,

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TABLE III
DIFFERENT NUMBERS OF FEMS

K_1	PSNR(dB)	BA[.%] QF 40	50	60	70	80	Average
1	43.74	97.62	99.19	99.77	99.97	100.00	99.31
2	43.94	97.63	99.27	99.69	99.93	99.99	99.30
3	44.20	97.95	99.32	99.73	99.89	99.98	99.37
4	44.49	97.94	99.48	99.86	99.98	99.99	**99.45**
5	**44.68**	97.65	99.26	99.68	99.87	99.96	99.28
6	43.98	97.28	99.08	99.69	99.97	99.99	99.20
7	42.57	96.24	98.12	98.92	99.44	99.75	98.49

TABLE IV
MODEL PERFORMANCE FOR DIFFERENT TRAINING EPOCHS

Epoch	PSNR(dB)	SSIM	BA[.%] QF 40	50	60	70	80	90	Average
20	42.41	0.9730	94.55	96.74	98.13	98.85	99.05	99.28	97.77
40	43.23	0.9728	96.25	98.52	99.34	99.73	99.87	99.92	98.94
60	43.35	0.9740	96.51	98.72	99.39	99.67	99.86	99.93	99.01
80	44.27	0.9772	97.55	99.12	99.69	99.89	99.97	99.97	99.36
100	**44.49**	0.9802	97.94	99.48	99.86	99.97	99.98	100	**99.54**
120	44.47	**0.9817**	97.48	99.08	99.73	99.95	99.99	100	99.37

λ_2 is set to 0.0005 and 0.00005, respectively. It is observed that when λ_2 is increased from 0.0001 to 0.0005, PSNR and SSIM are increased by 0.55dB and 0.0031, respectively, while BA is decreased by 0.97%. Meanwhile, when λ_2 is decreased from 0.0001 to 0.00005, PSNR and SSIM are decreased by 0.68dB and 0.0041, respectively, and BA is only increased by 0.05%. The above results show that λ_2 affects both watermarking invisibility and robustness as well due to the joint training of the encoder and the decoder. Therefore, considering the trade-off between invisibility and robustness, λ_1, λ_2 and λ_3 are set to 3, 0.0001 and 10, respectively.

In addition to the parameters of the loss function, K_1 also affects the watermarking performance. From Table III, we can find that PSNR is increased as K_1 is increased from 1 to 5, whereas when K_1 is continued to be grown, PSNR is declined. For watermarking robustness, when K_1 is 4, BA is the maximum. Thus, it denotes that suitable K_1 can increase the watermarking performance, and yet large K_1 causes over-communication between the image and watermark, which leads to some features suitable for watermarking being abandoned. Based on the above analysis, four FEMs are adaptive to compute cross-attention for enhancing image features, which help embed watermark into the image invisibly in the following SFM. Thus, we set K_1 to 4 for the optimal performance.

The number of training epochs also affects the watermarking performance, as shown in Table IV. From Table IV, we can see that watermarking invisibility and robustness are increased as training epochs grows. However, when the number of training epochs is 120, the watermarking performance is slightly decreased as PSNR and average BA are declined, which indicates that the model is already fitting at epoch 100. Therefore, to get the best model performance, we set the training epoch to 100.

TABLE V
PERFORMANCE OF 5-FOLD FOR DIFFERENT DATASETS[.%]

Datasets	Iterations	Identity	Crop (0.035)	Cropout (0.3)	Dropout (0.3)	Gaussian Blur (2)	JPEG (50)
COCO_fold	1	100	97.34	100	99.34	100	96.72
	2	100	96.62	100	99.17	100	96.80
	3	100	97.04	100	99.58	100	97.47
	4	100	96.41	100	99.26	100	97.88
	5	100	96.92	100	99.99	100	96.21
DIV2K_fold	1	100	96.79	100	99.71	100	97.11
	2	100	97.04	100	99.85	100	97.16
	3	100	96.93	100	99.94	100	96.89
	4	100	96.89	100	99.91	100	97.07
	5	100	96.76	100	99.93	100	97.18
ImageNet_fold	1	100	96.58	99.99	99.97	100	97.13
	2	100	96.42	99.99	99.85	100	97.16
	3	100	96.46	100	99.94	100	96.21
	4	100	96.61	100	99.95	100	97.11
	5	100	96.66	100	99.98	100	97.14

B. Embedding Capacity and Generalizability Discussion

Different embedding capacities also lead to different watermarking performances. We train three different models with different embedding capacities of 36 bits, 64 bits and 100 bits, respectively, and then tune them to the same image quality, as shown in Table VI. From Table VI, it can be seen that watermarking robustness is decreased as the embedding capacity is enlarged, especially on Crop (0.035) and JPEG (50) distortions. When the embedding capacity arrives at 100 bits, the average of BA is under 75%, which shows the resistance capability is degraded much. Thus, the proper embedding capacity can be acceptable for the proposed model, but it cannot work well as the embedding capacity is too large.

In order to demonstrate the generalizability of the proposed model, we perform 5-fold cross validation on the COCO [44], DIV2K [45], and ImageNet [46] datasets, respectively. Specifically, 10,000 images are selected randomly from the COCO [44] and ImageNet [46] datasets, namely, COCO_fold and ImageNet_fold, respectively. All images from the DIV2K [45], that is, 1,000 images, are used and named DIV2K_fold. Each dataset is segmented into five folds randomly, wherein four of the folds are used for training and the remaining one is used for testing. This iterative process is performed five times until all folds have been executed for one time testing. From Table V, it can be seen that in each dataset the proposed method achieves a good and similar performance for each iteration, and obtains similar robustness for different datasets. Moreover, the results of the five tests are averaged to obtain the final performance. From Table VI, we can see that BAs of COCO_fold are similar as those of COCO [44], which denotes that the proposed method has good generalization. From DIV2K_fold, ImageNet_fold, we still achieve good performance considering high BAs against various attacks, and it means the generalizability of the proposed model again.

Furthermore, cross-testing is performed on the DIV2K [45] and ImageNet [46] datasets, respectively, when the proposed model is trained on COCO [44]. From Table VI, compared with the results of COCO [44], similar performances on DIV2K [45] and ImageNet [46] are obtained. The above experimental results

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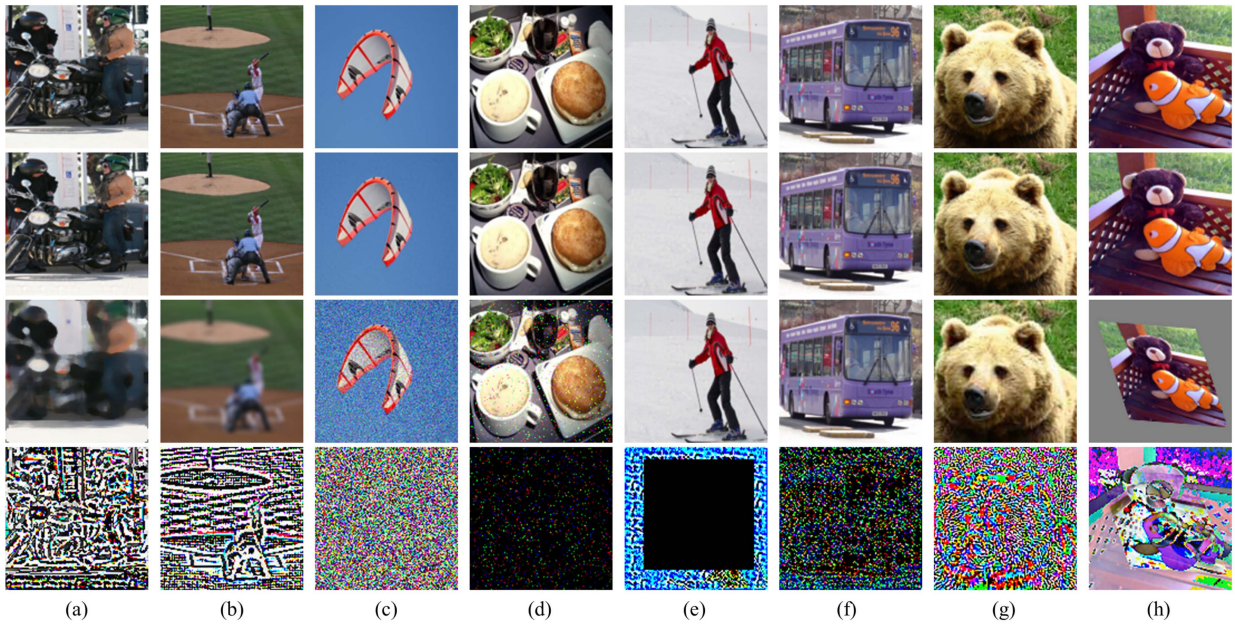


Fig. 5. Visualization of the watermarked image and different image processing distortion. From top to bottom: the cover image I_c, the encoded image I_e, the noised image I_n, the residual image $|I_e - I_n|$. (a) Median Blur (7), (b) Gaussian Blur (2), (c) Gaussian Noise (0.04), (d) Salt & Pepper (0.04), (e) Cropout (0.4), (f) Dropout (0.4), (g) JPEG (50), and (h) Affine attacks.

TABLE VI
PERFORMANCE WITH DIFFERENT EMBEDDING CAPACITIES AND DATASETS[.%]

Datasets	Watermark length	Identity	Crop (0.035)	Cropout (0.3)	Dropout (0.3)	Gaussian Blur (2)	JPEG (50)
COCO [44]	36	100	98.70	100	100	100	99.14
	64	100	97.17	100	100	100	97.73
	100	100	88.15	100	99.87	99.99	70.04
COCO_fold	64	100	96.86	100	99.47	100	97.02
DIV2K [45]	36	100	98.53	100	100	100	98.14
	64	100	96.97	100	99.95	100	97.31
	100	100	87.69	100	99.68	99.99	69.71
DIV2K_fold	64	100	96.88	100	99.87	100	97.08
ImageNet [46]	36	100	98.96	100	100	100	99.21
	64	100	97.01	100	100	100	97.98
	100	100	87.47	100	99.88	99.99	69.42
ImageNet_fold	64	100	96.54	99.99	99.93	100	97.15

TABLE VII
PSNR AND BA COMPARISONS FOR MEDIAN BLUR

Model	PSNR(dB)	BA[.%]			
		Windows			Average
		3×3	5×5	7×7	
HiDDen [13]	37.07	86.25	83.70	79.71	83.22
TSDL [15]	38.64	99.38	97.21	95.12	97.24
MBRS [17]	40.98	99.42	98.93	97.27	98.54
Fang [16]	32.5 ± 0.5	95.30	94.50	94.00	94.60
De-END [21]	41.18	99.81	99.61	98.34	99.25
Proposed	**44.76**	**99.93**	**99.85**	**99.55**	**99.78**

TABLE VIII
PSNR AND BA COMPARISONS FOR GAUSSIAN BLUR

Model	PSNR(dB)	BA[.%]				
		Variance				Average
		0.0001	0.5	1	2	
HiDDen [13]	46.21	95.44	95.21	94.33	84.37	92.34
TSDL [15]	45.97	99.92	99.79	98.48	93.21	97.85
MBRS [17]	47.91	98.64	98.25	97.66	87.80	95.59
Fang [16]	32.5 ± 0.5	–	90.40	92.10	92.00	91.50
De-END [21]	48.41	**100**	**100**	**99.51**	94.34	98.46
Proposed	**49.36**	98.90	98.96	99.01	**98.69**	**98.89**

prove that the proposed model has good generalization and can be applied in the real application. The subsequent experimental results are obtained on COCO [44].

C. Comparison of Specific Distortion Training

To test watermarking invisibility and robustness, we conduct experiments on different types of noises as illustrated in Fig. 5. From the residual image, we can find that each distortion changes the image much. For each noise, a specific network model is trained to show the learning ability of the proposed network model, and meanwhile, HiDDen [13], TSDL [15], MBRS [17], Fang [16], and De-END [21] are utilized to be compared.

1) Enhancement Techniques Attacks:

a) Median Blur: In the training stage, the window size is set to 7×7 for the training parameter. In the testing stage, we set the window sizes to 3×3, 5×5, and 7×7, respectively. Table VII shows that the proposed watermarking model is still the best in terms of visual quality and watermarking robustness. Especially for the image quality, the PSNR is at least 3 dB higher than those of other methods. Thus, the proposed model has the best capability of resisting median blur.

b) Gaussian Blur: We fix Gaussian blur with a variance of 2 in the training stage, and set the variance from 0.0001 to 2 in the testing stage. As found in Table VIII, the proposed watermarking model also obtains the highest image quality.

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TABLE IX
PSNR AND BA COMPARISONS FOR GAUSSIAN NOISE

Model	PSNR(dB)	BA[.%] Variance 0.01	0.02	0.03	0.04	0.05	Average
HiDDen [13]	36.25	89.58	86.46	83.96	83.12	79.17	84.46
TSDL [15]	39.46	92.08	91.25	88.33	87.08	82.92	88.33
MBRS [17]	39.70	99.91	99.42	98.10	96.09	94.15	97.53
Fang [16]	32.5 ± 0.5	90.50	–	–	–	–	90.50
De-END [21]	40.13	**100**	99.71	98.34	96.58	95.90	98.11
Proposed	**40.48**	**100**	**99.98**	**99.83**	**99.41**	**98.72**	**99.59**

TABLE X
PSNR AND BA COMPARISONS FOR SALT & PEPPER NOISE

Model	PSNR(dB)	BA[.%] Ratio 0.01	0.02	0.03	0.04	0.05	Average
HiDDen [13]	46.04	95.12	93.79	93.45	92.92	90.42	93.14
TSDL [15]	51.16	97.29	95.63	93.54	92.71	91.46	94.13
MBRS [17]	51.79	98.05	98.74	98.34	97.56	96.68	97.84
Fang [16]	32.5 ± 0.5	97.00	97.30	97.70	97.10	97.00	97.22
De-END [21]	52.43	99.41	99.51	99.22	99.12	98.73	99.20
Proposed	**52.71**	**99.90**	**99.83**	**99.74**	**99.52**	**99.22**	**99.64**

Robustness on Gaussian blur is higher than those of HiDDen [13], TSDL [15], MBRS [17], and Fang [16]. Compared with De-END [21], the proposed model is lower for the variances of 0.0001, 0.5, and 1, but higher for the variance of 2. Considering the average BA, the proposed model performs better under the high distortion conditions. It shows that the proposed model has the good capability of resisting strong distortions.

2) Noise Addition Attacks:

a) Gaussian Noise: Gaussian noise is a kind of noise whose density follows Gaussian distribution. In the training stage, the variance is set from 0.001 to 0.04, and the testing variance is changed from 0.01 to 0.05. Table IX shows that the proposed model is the highest, which represents that the image quality is the best, and robustness on Gaussian Noise with different variances is the strongest.

b) Salt & Pepper Noise: We randomly select the training ratio of Salt & Pepper noise from 0.001 to 0.04, and set the testing ratio from 0.01 to 0.05. As observed in Table X, the proposed WFormer obtains the highest PSNR as well. BAs are higher than those of other compared models for different densities, which means that the proposed watermarking model is robust to Salt & Pepper Noise.

3) Geometric Attacks:

a) Cropout: Cropout means that a certain ratio of the image portion is preserved, and the other region is replaced by the same region of the cover image. In the training stage, the corresponding ratio is 60%. For the testing, the ratio is set from 50% to 90% as shown in Table XI. Compared with the other five watermarking models, the PSNR of the proposed watermarking model is the best. The proposed watermarking model achieves at least 1.5dB gain compared to De-END [21], which expresses the proposed watermarking model obtains the high quality of the encoded image. For watermarking robustness, the average BA of the proposed model for different ratios is higher than those of other watermarking models, which indicates the proposed model can resist Cropout well. Especially for high distortion, when the ratio is 50%, the corresponding BA is at least 5.0% higher than those of other models, which means the proposed watermarking model has the ability to resist high distortions.

TABLE XI
PSNR AND BA COMPARISONS FOR CROPOUT

Model	PSNR(dB)	BA[.%] Ratio 90%	80%	70%	60%	50%	Average
HiDDen [13]	40.62	95.63	94.73	88.75	76.88	61.67	83.53
TSDL [15]	47.48	98.72	98.54	96.88	93.75	93.21	96.22
MBRS [17]	48.05	99.71	99.22	97.18	90.43	83.50	94.01
Fang [16]	32.5 ± 0.5	98.30	97.90	97.10	95.30	–	97.15
De-END [21]	50.74	**100**	**100**	99.51	97.28	91.21	97.60
Proposed	**52.72**	99.99	99.98	**99.97**	**99.96**	**98.35**	**99.65**

TABLE XII
PSNR AND BA COMPARISONS FOR DROPOUT

Model	PSNR(dB)	BA[.%] Ratio 80%	70%	60%	50%	40%	Average
HiDDen [13]	42.59	90.21	89.58	87.08	86.74	82.71	87.26
TSDL [15]	53.59	97.54	95.21	93.54	92.29	90.42	93.80
MBRS [17]	58.63	96.29	94.73	94.15	92.58	90.63	93.68
Fang [16]	32.5 ± 0.5	97.40	–	–	–	–	97.40
De-END [21]	58.97	**100**	**100**	**100**	**99.51**	94.63	**98.83**
Proposed	**58.99**	99.59	99.22	98.68	97.66	**95.84**	98.19

TABLE XIII
PSNR AND BA COMPARISONS FOR JPEG COMPRESSION

Model	PSNR(dB)	BA[.%] QF 40	50	60	70	80	90	Average
HiDDen [13]	33.29	86.67	91.24	92.92	93.33	93.54	94.38	92.01
TSDL [15]	39.39	91.04	91.46	93.96	94.21	94.35	94.74	93.29
MBRS [17]	45.16	94.83	94.93	96.68	97.66	97.66	98.84	96.77
Fang [16]	32.5 ± 0.5	–	91.50	92.50	93.70	94.30	95.00	93.40
De-END [21]	**46.84**	**98.15**	**99.02**	**100**	**100**	**100**	**100**	**99.53**
Proposed	45.41	95.83	98.79	99.60	99.92	99.96	**100**	99.02

b) Dropout: Dropout randomly preserves a certain ratio of image pixels, and the left pixels are replaced by the cover image. Different from Cropout, it selects image pixels randomly. For training, the ratio is set to 60%, and then the ratio is assigned from 40% to 80% for testing as shown in Table XII. Similar to Cropout, the PSNR of the proposed model is the highest among all compared models for Dropout. For watermarking robustness, De-END [21] is a little better, and the proposed model is the second. However, for the high distortion, such as the ratio of 40%, BA of the proposed model is 1.0% higher than that of De-END [21], which also proves robustness on the high strength of image attacks.

4) Other Attacks:

a) JPEG Compression: JPEG is a non-differential distortion and always takes place in image saving or format conversion. In the training stage, simulated JPEG compression proposed by MBRS [17] is used in the noise layer, and the quality factor (QF) is 50. For testing, QF is set from 40 to 90 to show robustness as shown in Table XIII. De-END [21] obtains the highest PSNR,

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TABLE XIV
PSNR AND BA COMPARISONS FOR AFFINE ATTACKS

Model	PSNR(dB)	BA[.%] Degree, Translate, Scale, Shear				
		(−10°,10°), (0.1,0.1), (0.7,0.7), (−30°,30°)	(−0°,0°), (0.2,0.2), (0.7,0.7), (−30°,30°)	(−0°,0°), (0.1,0.1), (0.6,0.6), (−30°,30°)	(−0°,0°), (0.1,0.1), (0.7,0.7), (−20°,20°)	Average
HiDDen [13]	33.58	70.03	68.72	73.19	77.88	72.46
MBRS [17]	35.01	88.39	81.62	84.35	92.11	86.62
De-END [21]	34.26	69.16	67.86	70.14	73.22	70.09
Proposed	**36.91**	**91.79**	**89.63**	**93.32**	**100**	**93.68**

and the highest BAs for JPEG compression with the different values of QF. However, the proposed model also ranks the second place, and is better than other models except for De-END [21]. Considering such PSNR and BA levels, the proposed model still has the high image quality and strong robustness.

b) Affine attacks: The affine attacks include degrees, translate, scale and shear. In the training phase, the affine attack is set as degrees = (−0°, 0°), translate = (0.1, 0.1), scale = (0.7, 0.7), and shear = (−30°, 30°). For testing, four different strengths of affine attacks are compared, as shown in Table XIV, and obviously the proposed model achieves the best results in terms of invisibility and robustness. It can be clearly seen that the effect of shear on robustness is the largest, because shear can severely damage the distribution of images, which destroys the watermark information and reduces robustness.

Based on the above experimental results, we can conclude that for resisting Cropout, Gaussian Noise, Salt & Pepper noise, Median blur, and Affine attacks with different strengths, the proposed model is the best. Only BAs of the proposed model are little lower than those of De-END [21] for Dropout and Gaussian blur of low strengths, and JPEG compression. It is mainly because De-END [21] adopts a decoder-guided approach, in which only a few latent features are extracted for embedding watermark. During training a specific attack with low distortion, features can be easily mined. However, when the distortion strength is increased, the feature number is not enough to satisfy for resisting corresponding attacks, such as Gaussian Blur(2) and Dropout(0.4). By comprehensive consideration of test results for resisting all image attacks, the proposed model is superior to the other five watermarking models. There two main reasons for supporting the performance of the proposed model. Firstly, the watermark preprocess module (WPM), the feature enhancement module (FEM), and the soft fusion module (SFM) are integrated into the encoder to assist the watermark embedding process. The WPM expands watermark to generate valid watermark features for supplying the encoding process by reducing the influence of embedding watermark into the image, so that the image quality is improved. The FEM adjusts image features to be adaptive to watermark features by capturing cross-feature dependencies between the image and watermark from different domains, which is good preparation for watermark fusion. The SFM allows communication to build their correlations by computing self-attention and cross-attention, so that the image is fused with watermark well for obtaining high robustness and

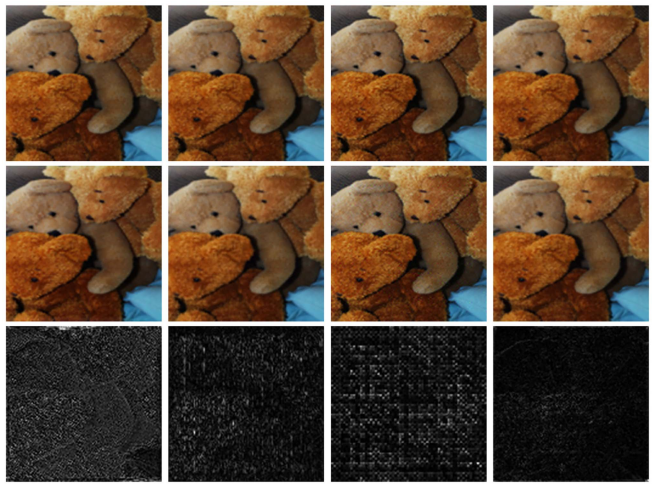


Fig. 6. Visualization comparison under Crop attack training. From top to bottom are cover image, encoded image and residual image (normalization). (a) HiDDen [13], (b) MBRS [17], (c) De-END [21], and (d) Proposed.

invisibility. Secondly, in the watermark extraction process, the decoder is designed with a network structure symmetrical to the encoder, which performs the down-sampling operation to extract the watermark information. Since the encoder and the decoder are jointly trained, this symmetric design helps to improve the coupling between the encoder and the decoder, thus enhancing the accuracy of watermark extraction.

5) Imperceptibility Analysis: As illustrated in Fig. 5, the encoded image and the cover image are visually similar, and the human eye cannot observe the difference, which indicates that the proposed model is able to imperceptibly embed watermark into the cover image and maintain the high visual quality. To further demonstrate invisibility of the proposed model, a visual comparison with HiDDen [13], MBRS [17] and De-END [21] is performed as illustrated in Fig. 6. Fig. 6 shows the encoded images and corresponding residual images with normalization [17]. The residual image can represent watermark distribution and corresponding embedded capacity. It can be clearly found that, a large number of watermark bits exist in the encoded image for HiDDen [13], which causes low invisibility. The main reason is that watermark is simply duplicated and nearly each image pixel holds watermark bits, so over-redundancy is generated to degrade the image quality. For MBRS [17], the embedded watermark is still obvious. Although MBRS [17] uses convolution operations to extend watermark instead of simple duplication, convolution operations have the perceptual field limitation, which can only capture local watermark features but not all effective features, thereby degrading the image quality as well. Similarly, De-END [21] also has a large number of embedded watermark bits distributed over the encoded image, and especially the encoded image has noises as illustrated in Fig. 6(c). On the contrary, the proposed model has fewer watermark bits but can still has the capability of robustness on different image attacks. The main reason is that the WPM can model long-range relationships to capture the global watermark features. This is

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TABLE XV
BA COMPARISONS FOR SIX DISTORTIONS[.%]

Model	Watermark length	Identity	Crop(0.035)	Cropout(0.3)	Dropout(0.3)	Gaussian Blur(2)	JEPG(50)
HiDDen [13]	30	100	88.00	94.00	93.00	96.00	63.00
ReDMark [14]	30	100	**100**	92.50	92.00	50.00	76.40
TSDL [15]	30	100	89.00	97.30	97.40	98.60	76.20
DA [18]	30	100	81.15	78.57	77.13	92.80	82.83
MBRS [17]	30	100	95.85	**100**	99.99	99.99	95.52
De-END [21]	**64**	100	64.17	99.21	99.95	88.93	81.89
Proposed	36	100	98.70	**100**	**100**	**100**	**99.14**
Proposed	**64**	100	97.17	**100**	**100**	**100**	**97.73**

able to reduce ineffective and redundant watermark information to be embedded, thus improving watermarking invisibility. Meanwhile, the FEM and the SFM allow the interaction of the image and watermark to find inconspicuous image features for holding watermark to improve watermarking invisibility.

TABLE XVI
COMPANIONS WITH CIN[20][.%]

Model	Message length	PSNR(dB)	Crop(0.035)	Cropout(0.3)	JEPG(50)
CIN [20]	64	34.22	86.60	86.73	93.23
Proposed	64	**35.44**	**96.67**	**100**	**94.55**

D. Comparison of Combined Distortion Training

In real transmission, the image may be usually under different attacks, and hence the excellent watermarking model can resist various image noises rather than only one unique attack. Therefore, we train the model by using combined distortions. Specifically, the noise layer includes JPEG-Mask (50), real JPEG (10), Identity, Crop (0.0225), Cropout (0.2) and Gaussian Blur (2), and each training batch randomly selects one of the distortions. HiDDen [13], ReDMark [14], TSDL [15], DA [18], MBRS [17], and De-END [21] are used to be compared. To be fair comparisons, we test six distortions under the condition that corresponding PSNRs of watermarked images are around 33.5dB as shown in Table XV, wherein Identity means that the image is not attacked.

As shown in Table XV, when the watermark length is 36 bits, only BA of the proposed watermarking model for Crop (0.035) is a little lower than that of ReDMark [14], and yet the proposed watermarking model has the highest BA for other distortions among all compared watermarking models. In particular, BAs for Cropout (0.3), Dropout (0.3), and Gaussian blur (2) are all 100%, which denotes watermark can be completely extracted. BA of the proposed model is 98.70% for Crop (0.035), which is at least 2.8% higher than those of other models, except ReDMark [14]. For JPEG (50), BA of the proposed model is 99.14%, which is at least 3.0% higher than those of other models. Since other models only embed 30 bits of watermark into the image, except De-END [21], the embedding capacity of the proposed model is also higher. Thus, our watermarking model has achieved the best performance in terms of watermarking capacity and robustness. When watermarking capacity is increased to 64 bits, robustness is decreased as shown in Table XV in contrast to 36 bits of watermark. However, the proposed model is still superior to the other six models overall considering the capability of resistance to image attacks. Among them, De-END [21] has shown a good performance in some specific distortions, but is not satisfactory for combined distortions. The main reason is that to resist combined attacks, large number of latent features should be prepared for embedding watermark, and yet features from De-END [21] are no sufficient to hold watermark. The above results prove the effectiveness of the proposed model in the case of low and large watermarking capacities. In real applications, appropriate watermarking capacity can be adjusted for different user requirements.

In order to further prove robustness, five noises with different intensities are tested for comparisons as illustrated in Fig. 7. We also use two trained models of WFormer for embedding 36 bits and 64 bits, namely WFormer($L = 36$) and WFormer($L = 64$), respectively. It is clearly seen that WFormer($L = 36$) is better than WFormer($L = 64$) since large embedding capacity will reduce robustness. However, it's good to see that both WFormer($L = 36$) and WFormer($L = 64$) have better performance than HiDDen [13] and MBRS [17]. Especially for Gaussian blur distortion, the proposed watermarking model obtains almost 100% of BA for different variances, which is significantly higher than those of other models. The above comparison results prove that the proposed model has a better generalization performance against different intensities of distortions.

We also compared CIN [20] with the same watermarking capacity as shown in Table XVI. Obviously, the proposed watermarking model achieves better performance in terms of the image visual quality and robustness. To be more concrete, the proposed model has at least 1.0 dB gain over CIN [20]. For Crop(0.035), BA of the proposed model is at least 10.0% higher than that of CIN [20]. For Cropout(0.3), the corresponding BA is 100%, which is at least 13.0% higher than that of CIN [20]. For JPEG (50), the proposed model has at least 1.0% BA boost over CIN [20]. Thus, the effectiveness of the proposed model is demonstrated again. Based on the above comparison results with existing DNN based watermarking models, the proposed WFormer is superior in the light of watermarking invisibility, robustness and capacity totally. The following analyzes the reasons for the superiority.

Firstly, none of the above compared models consider the watermark information when extracting image features, which results in some fragile features extracted for watermark fusion. When watermark is fused with those features, robustness is decreased. However, the proposed model is able to capture

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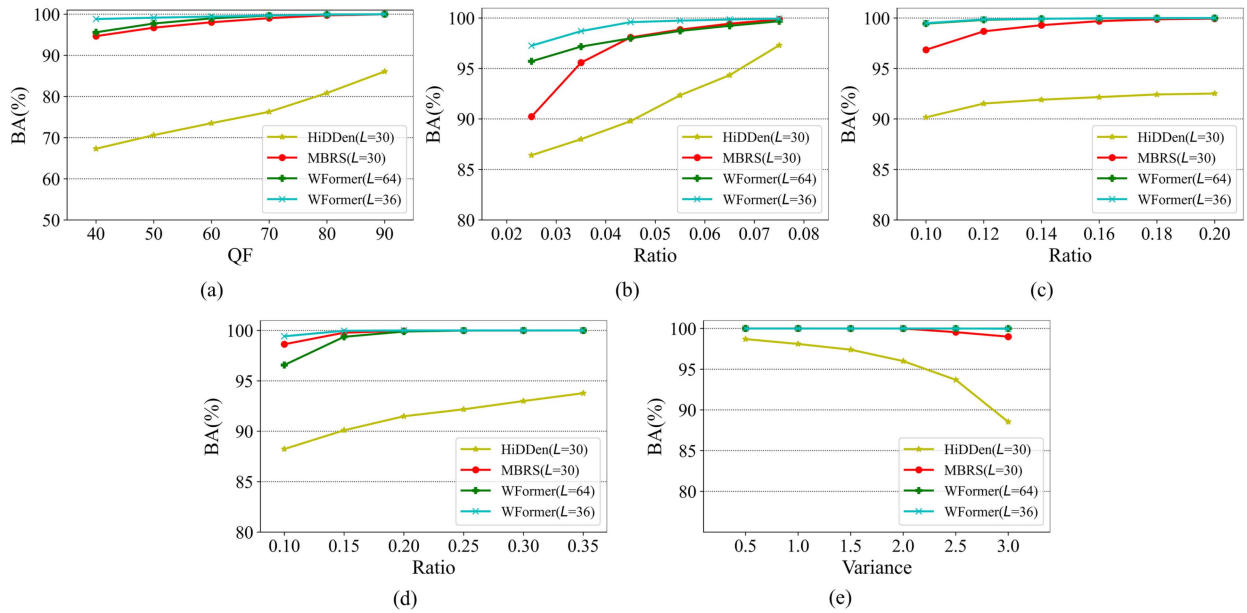


Fig. 7. Robustness comparison on different noises. (a) JPEG Compression, (b) Crop, (c) Cropout, (d) Dropout, and (e) Gaussian Blur.

the correlation between the image and watermark by computing cross-attention, which enhances robust image features and suppresses fragile image features for increasing robustness. Secondly, all compared models use channel concatenation for watermark fusion without any communication between watermark and image features, which leads to watermark fused with fragile features. In contrast, the proposed model is able to allow the image and watermark to communicate extensively by calculating cross-attention, so that robust image features are mined to improve watermarking robustness.

E. Comparison of Untrained Distortions

In order to demonstrate the robust generalization ability of the proposed watermarking model, we conduct experiments to resist unknown distortions. We pre-train the watermarking model with a specific distortion, and then other untrained distortions are used for testing. Fang [16] is used to be compared and corresponding PSNR is adjusted to be similar to that of the proposed watermarking model for evaluating robustness as illustrated in Fig. 8. From Fig. 8, it is clearly found that the proposed watermarking model is better than Fang [16] for most of the distortions. Only for resisting JPEG compression, BAs of the proposed watermarking model are less when Gaussian Noise and Salt & Pepper are the trained distortions, respectively. The main reason is that Fang [16] uses mask-guided frequency enhancement in the second stage of training, which will guide watermark to embed in the low frequency region, while the JPEG compression mainly compresses the high frequency region of the image, so that robustness is better for JPEG compression. However, considering the average BA, the proposed watermarking model is superior to Fang [16], which demonstrates that we have the generalization ability to resist different unknown attacks.

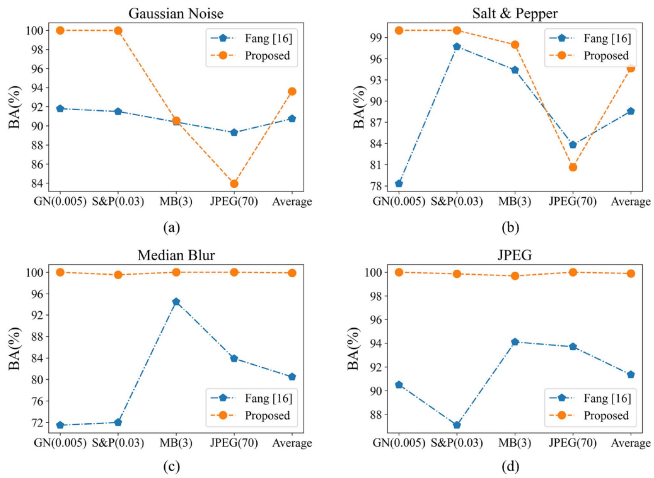


Fig. 8. Comparison of untrained distortions when different specific noises are trained. (a) Gaussian Noise (GN), (b) Salt & Pepper (SP), (c) Medium Blur (MB), and (d) JPEG Compression.

Furthermore, the watermarking model is pre-trained by combined distortions, and then tested on other untrained distortions, including Median Filter, Grid Crop, Adjust Hue, Adjust Saturation, and WebP. For each distortion, three different distortion strengths are tested and compared with HiDDen [13], MBRS [17] and De-END [21], as shown in Table XVII. From Table XVII, the proposed model is robust to the different distortions since the average values of BA is more than 93%. Compared to the second best model, the BA gain is more than 3%. Moreover, the average of NC is 0.87, and also 0.05 higher than that of MBRS, which proves robustness again. Thus, The proposed model achieves the best performance among all compared models. It also shows the robustness generalizability of the proposed

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TABLE XVII
ROBUSTNESS COMPARISON OF UNTRAINED DISTORTIONS

Distortion	HiDDen [13]		MBRS [17]		De-END [21]		Proposed	
	BA	NC	BA	NC	BA	NC	BA	NC
Median Filter (5×5)	72.58	0.6191	**99.96**	**0.9994**	91.06	0.8305	98.06	0.9624
Median Filter (7×7)	64.28	0.4352	88.01	0.7659	73.45	0.6318	**95.68**	**0.9263**
Median Filter (9×9)	59.76	0.3984	58.84	0.3736	54.83	0.3217	**91.12**	**0.8541**
Grid Crop (0.7)	73.02	0.5875	99.93	0.9989	95.49	0.9093	**100.00**	**1.0000**
Grid Crop (0.8)	65.33	0.4414	99.76	0.9963	89.09	0.7925	**99.97**	**0.9996**
Grid Crop (0.9)	59.40	0.3909	97.99	0.9606	76.63	0.6622	**98.89**	**0.9790**
Adjust Hue (0.44)	68.34	0.5013	95.44	0.9106	91.53	0.8386	**95.68**	**0.9122**
Adjust Hue (0.46)	64.10	0.4394	86.46	0.7397	80.32	0.6832	**88.27**	**0.7675**
Adjust Hue (0.48)	58.20	0.3987	67.33	0.4860	62.60	0.4081	**74.59**	**0.6341**
Adjust Saturation(5.0)	79.09	0.6804	99.86	0.9943	97.01	0.9406	**99.92**	**0.9951**
Adjust Saturation(10.0)	77.59	0.6677	99.77	0.9939	96.43	0.9316	**99.83**	**0.9945**
Adjust Saturation(15.0)	76.36	0.6493	99.63	0.9928	96.19	0.9257	**99.69**	**0.9938**
WebP (20)	53.71	0.3268	79.77	0.6781	65.64	0.4588	**81.60**	**0.6959**
WebP (30)	55.08	0.3456	84.87	0.7125	70.83	0.5627	**85.10**	**0.7195**
WebP (40)	55.79	0.3544	**88.92**	**0.7784**	75.17	0.6496	88.16	0.7592
Average	65.51	0.4824	89.77	0.8254	81.08	0.7031	**93.10**	**0.8795**

model, since those distortions are untrained but still can be resisted. As above stated, the WPM, the FEM and the SFM help to obtain the good watermarking performance, and in the following ablation study, effectiveness of these three modules will be demonstrated.

F. Ablation Study

1) Influence of WPM: At first, we will discuss the influence of the WPM on the proposed WFormer. In the WPM, Transformers are employed to learn self-attention for expanding watermark, which generates valid watermark redundancies for robustness. In order to prove the effectiveness of the WPM, it is replaced by simple duplication in HiDDen [13], watermark preprocess of MBRS [17], and convolution blocks [49] for testing, namely, H_WFormer, M_WFormer, and C_WFormer, respectively. We use JPEG compression for training and testing, and the corresponding test results are shown in Table XVIII.

Table XVIII shows that the proposed watermarking model is the best, which proves the WPM based on Transformer plays an important role in watermarking performance. Obviously, H_WFormer performs the worst, and the main reason is that the simple watermark duplication operation of HiDDen [13]

TABLE XVIII
PSNR AND BA COMPARISONS OF DIFFERENT WATERMARK EXPANSION TECHNIQUES

Model	PSNR(dB)	BA[.%]					
		QF					Average
		40	50	60	70	80	
H_WFormer	33.38	68.70	70.93	72.33	73.80	75.80	72.31
M_WFormer	43.83	98.08	99.37	99.73	99.88	99.95	99.41
C_WFormer	43.53	97.82	99.18	99.71	99.95	99.99	99.33
Proposed	**44.49**	97.94	99.48	99.86	99.98	99.99	**99.45**

TABLE XIX
PSNR AND BA COMPARISONS OF EXTRACTING DIFFERENT IMAGE FEATURES

Model	PSNR(dB)	BA[.%]					
		QF					Average
		40	50	60	70	80	
T_WFormer	43.33	97.69	99.03	99.74	99.91	99.97	99.27
Q_WFormer	43.70	97.86	99.42	99.79	99.95	99.98	99.40
Proposed	**44.49**	97.94	99.48	99.86	99.98	99.99	**99.45**

for watermark expansion leads to watermark over-redundancy, which causes the lowest PSNR. In addition, the average BA of H_WFormer is much lower than that of the proposed model, because the simple watermark duplication operation does not expand watermark efficiently and the expanded watermark is not valid enough for resisting image noises. Compared with M_WFormer and C_WFormer, BA of the proposed model is a little higher, PSNR is at least 0.5dB higher. It is mainly because the convolution operation of the watermark preprocess in MBRS and convolution blocks have the defects of the limited receptive field, which results in incomplete watermark features embedded into the image so that the corresponding performance is degraded. On the contrary, the Transformer Block can extract intact watermark features in the process of watermark expansion, since it has the global modeling ability, which increases watermarking invisibility and robustness.

2) Influence of FEM: The FEM is used to extract effective features for fusing with watermark, and is discussed to show its effectiveness. At first, The FEM is replaced by the Transformer Block [37] in WFormer, namely, T_WFormer. Secondly, in the FEM, Query(Q_c) and Key(K_w) are computed to replace Q_w and K_c, respectively, namely Q_WFormer. Those two watermarking models are compared with the proposed model as shown in Table XIX. Compared to T_WFormer, the average BA of the proposed watermarking model is higher, and the corresponding PSNR is at least 1.0 dB higher. It indicates the importance of communication between the image and watermark for extracting image features. For T_WFormer, the Transformer Block only computes self-attention of the cover image, the extracted image features lack cross-feature dependencies, which brings about the poor fusion between the image and watermark. For the proposed model, besides self-attention, cross-attention between them is computed to adjust image features toward watermark features as the preparation for better watermark fusion.

Moreover, in contrast to Q_WFormer, the proposed watermarking model is better in terms of PSNR and BA as well. Although for both two watermarking models, communication is

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TABLE XX
PSNR AND BA COMPARISONS FOR DIFFERENT FUSION

Model	PSNR(dB)	BA[.%]					
		QF					Average
		40	50	60	70	80	
D_WFormer	43.38	96.77	98.79	99.52	99.84	99.94	98.97
CW_WFormer	44.16	97.55	99.16	99.72	99.90	99.98	99.26
CI_WFormer	43.77	97.49	99.15	99.73	99.93	99.98	99.26
Proposed	**44.49**	97.94	99.48	99.86	99.98	99.99	**99.45**

TABLE XXI
ANALYSIS OF COMPUTATIONAL COST

Model	Params [M]	FLOPs [G]	Speed [im/s]
HiDDeN [13]	0.40	3.52	62.93
MBRS [17]	5.80	13.36	17.67
De-END [21]	0.41	3.91	18.38
WFormer	1.72	13.83	14.29

TABLE XXII
COMPARISON OF COMPUTATIONAL COSTS OF WFORMER WITH THREE VARIANTS

Model	Params [M]	FLOPs [G]	Speed [im/s]
H_WFormer	0.81	10.10	19.23
T_WFormer	1.72	12.91	15.87
D_WFormer	1.66	12.62	16.13
WFormer	1.72	13.83	14.29

built between the image and watermark, they are different for capturing correlations. In the proposed watermarking model, each token of watermark is correlated with tokens of the cover image to find suitable image regions for embedding, such as inconspicuous and texture regions, whereas in Q_WFormer each token of the cover image is related to tokens of watermark to find suitable watermark bits to be embedded into each region. As we know, not every region of the cover image is suitable to embed watermark for invisibility and robustness, and yet all regions of the image can be supplied for embedding watermark bits in Q_WFormer, which degrades the watermarking performance. Therefore, it is worth noting that the FEM acquires the relationships between the cover image and watermark, and is properly designed for extracting and enhancing the main latent image features.

3) Influence of SFM: The SFM is used to compute self-attention and cross-attention on tokens from the image and watermark to embed watermark into the image. To investigate the influence of the SFM, three watermarking models are designed. In the first model, the SFM is replaced by direct fusion, that is, watermark feature channels are directly concatenated with image feature channels, namely D_WFormer. In the second model CW_WFormer, cross-attention for watermark features is removed from the SFM, that is, only self-attention of watermark features is computed and self-attention and cross-attention for image features are computed. In the third model CI_WFormer, cross-attention for image features is deleted, that is, only self-attention of image features is calculated and mixed attentions of watermark features are still preserved.

Table XX shows that PSNR of the proposed watermarking model is at least 1.0 dB higher, and robustness on JPEG compression with different QFs is stronger in contrast to D_WFormer. It denotes that direct fusion will damage the image a lot since two different modalities are not communicated with each other well. In addition, without self-attention and cross-attention computation, watermark cannot be merged well with image features for robustness yet. The SFM permits communication between two different modalities by capturing the cross-feature dependencies to obtain better fusion. More importantly, watermark features can be adapted to the distribution of image features for high image quality and capability of resistance to image noises. When the cross-attention computation is removed from the SFM, it is easily seen that the watermarking performance is decreased as well since CW_WFormer and CI_WFormer are worse than the proposed watermarking model as shown in Table XX. Thus, it is concluded that whether they are watermark features or image features, computing cross-attention helps to communicate between them for improving watermarking performance. In addition, self-attention also helps to extract watermark and image features for fusion since CW_WFormer and CI_WFormer are better than WS_Former. It is easily remarked that applying self-attention and cross-attention ensures good watermarking performance.

G. Comparison of Computational Costs

In order to evaluate the computational cost of the proposed model, the model size, time complexity and throughput are utilized to be compared with HiDDen [13], MBRS [17] and De-END [21] as shown in Table XXI. Here, the model size means the number of parameters, and time complexity and throughput are represented by floating-point operations per second (FLOPs) and the number of images processed per second, respectively.

From Table XXI, in terms of parameters, the proposed model is larger than HiDDen [13] and De-END [21], but smaller than MBRS [17]. The main reason is that HiDDen [13] uses simple convolutional operations and De-END [21] shares the parameters of two decoders for reducing the model size. However, MBRS [17] employs a large number of attention modules so that the corresponding model size is enlarged as well. Considering FLOPs and Speed, the other three models are better than the proposed model. The main reason is that the proposed model includes self-attention module, which has the complex structure for running, thereby resulting in large FLOPs and the low Speed. In all, although the proposed model has larger time complexity and smaller throughput, has better invisibility and higher robustness. It can be acceptable to sacrifice some computational cost while ensuring the watermarking performance. Moreover, the parameters of the proposed model are relatively small, and only takes up 1.72M, which means that the proposed model can be deployed in real application scenarios.

Further, we show the computational cost comparison of the proposed WFormer with the three variants in Table XXII. It is observed that H_WFormer has the smallest computational cost, which indicates that the WFM has the greatest impact on the computational cost of Wformer. However, Table XVIII

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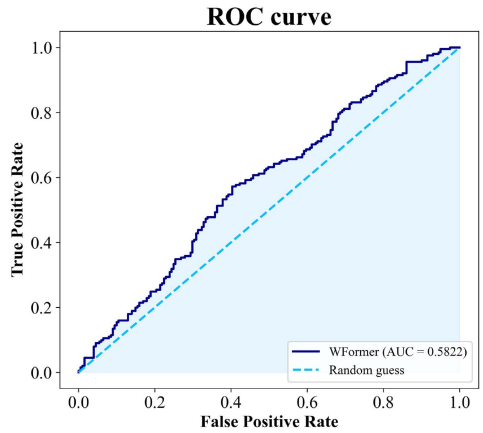


Fig. 9. ROC curve produced by StegExpose [50] for the proposed WFormer

illustrates that H_WFormer affects the watermarking performance much. Therefore, the WFM is necessary to the proposed WFormer although increases the computational time much. In addition, T_WFormer, D_WFormer and WFormer have similar computational costs, which indicates that FEM and the SFM do not significantly increase the computational cost. Since the FEM and the SFM can improve the watermarking performance as shown in Tables XIX and XX, these two modules are deployed without increasing the run time too much.

H. Security Analysis and Limitations

We use an open-source testing tool, called StegExpose [50], to test the security of the proposed model, that is, evaluating difficulty of detecting watermark embedded in the cover image. Specifically, we randomly select 2000 cover images from COCO [44]. StegExpose [50] outputs a probability to distinguish the cover image from the watermarked image. When the probability is 0.5, it indicates a random guess. If the probability is large, it means that watermark can be detected. Fig. 9 shows the receiver operating characteristic (ROC) curve of the proposed WFormer. We can see that the value of the area under the ROC curve (AUC) is 0.5822, which is close to 0.5. It indicates that the embedded watermark information cannot be detected easily, and the proposed model has some security.

Based on above analysis, although the proposed WFormer can resist most of the distortions well and have security, there are still some limitations that need to be improved in future work.

1) Robustness against real JPEG compression is not strong enough. In the noise layer, although the mini-batch of real and simulated JPEG compression proposed by MBRS [17] are used, it is still different from the real JPEG compression. How to train the non-differentiable noises is important for robustness.

2) Training of the proposed model is time-consuming. Since Transformers have complex structure in the WFM, have high time complexity in calculating self-attention and cross-attention for increasing the time cost. Designing a low-light attention module to take place of the WFM is critical.

How to optimize the model performance from the above two aspects is the focus of the future work.

V. Conclusion

In this paper, we have presented a Transformer based soft fusion for robust image watermarking, namely WFormer. The motivation comes from the main drawbacks of the existing deep neural network (DNN) based image watermarking models, that is, over-redundancies generated by simple duplication for watermark expansion, and communication between the cover image and watermark is omitted in the processes of image feature extraction and direct fusion. To produce effective watermark features, the watermark preprocess module (WPM) is deployed to expand watermark by computing its self-attention. To enhance valid image features toward watermark features for helping better fusion, a feature enhancement module (FEM) builds communication between them by acquiring cross-feature dependencies. More importantly, instead of direct fusion, the soft fusion module (SFM) is designed by utilizing Transformers to bridge the gap between two different modalities. Specifically, besides obtaining their own self-attention, cross-attention is calculated to embed watermark by constructing their relationships. Experimental results prove that the proposed WFormer is superior to the existing state-of-the-art watermarking models in terms of invisibility, robustness, and embedding capacity. However, the proposed model still has low robustness on the real JPEG compression, and is time consuming. In the future work, we will try to fix them.

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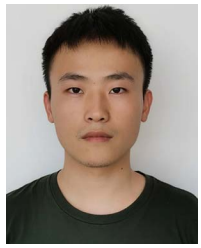
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